

2.3 Satellite Phones

A Little background

What happens when you take your antenna into outer space? You get a cell that covers an area hundreds of kilometres wide, and the whole world becomes your cellular network. This is precisely what the Iridium project did, when 66 satellites were launched into orbit with the hope of creating a mobile network that would reach every corner of the globe. It didn't catch on as well as they'd hoped, though. The jet-setting executives (for whom these phones were meant) didn't quite fancy the handsets, which were about the size of the old '80s bricks. The signal from a satellite did reach all corners of the globe, but only *outdoors*. Inside a building (where a jet-setting executive is more likely to be spotted), the signal quality took a severe hit. On top of all this, the decreasing cost and increasing popularity of GSM quickly paved the way to Iridium's bankruptcy.

Iridium's satellites, however, still orbit the earth and are used by other companies to provide satellite telephone services. Also playing the field are GlobalStar, Inmarsat, ACeS and Thuraya. Each of these providers uses a different technology for their services.

How The Satellite Phone Call Works

A satellite 'constellation' consists of many *Low Earth Orbiting (LEO)* satellites. When you switch on your satellite phone, it sends a signal straight up to the nearest satellite. The satellite then relays this signal to the nearest *Land Earth Station (LES)*, registering you with the constellation (a similar concept to registering on a cellular network). If you are calling someone at the other end of the world, your signal will be bounced off many satellites in the constellation before it is beamed down to the recipient.

Commentary On The Above

Satellite phone service providers offer common Internet services through their own gateways and outgoing servers, which is good, because one doesn't then have to deal with the eccentricities of